

selecting raw materials. In his *Pictorial Handbook for Making Porcelain* (《陶冶圖說》), Tang Ying recorded in the chapter of *Quarrying and Clay-Preparing*: choosing the Qimen porcelain stone with pure and fine colour and the kaoline clay with its quality as clam powder and colour white dotted with shiny spots as the main raw materials.

The raw stone was quarried from mines and then pulverised and washed and purified, made into brick-like clay lumps which were called "Dunzi (Clay Brick)" in Jingdezhen dialect. According to the description in the *Pictorial Handbook for Making Porcelain*, the washing and purifying method was to steep the clay in water vat, to stir and mix them with wood-sticks, to take dregs out, to filter with yarn sack so as to form thick slurry, then to wrap them with fine cloth and put them in bottomless wooden casket with new bricks underneath and bricks atop, to press and absorb water, to transfer them onto slab-stones and to turn them up and down with spades so as to make the clay lumps sturdy. After one and another complicate processing links, the purified raw materials were refined and then burned into deliberate and exquisite bodies of porcelains.

The *Qinghua* pigment should be burned in kiln before being classified into superior, medium and inferior levels according to colour and quality, and pulverised into small particles in mortar and applied to porcelain bodies of different grades. Thus the colour tone of the top-quality *Qinghua* porcelains of Kangxi Reign was transparent and limpid.

REASONABLE FORMULA Potters not only had clear idea of the functions of every kind of pigments but commanded the reasonable formula to make body and glaze slip after repeated experiments. The mud (clay) used to make body came from the composition of kaoline clay and porcelain stones, and the potters determined the ingredients of the formula in line with the shape, size, thickness of the product and firing-temperature. Usually the shape of the product with more content of kaoline clay would not easily change while being burned but need a rather higher firing-temperature. The glaze slip was made by steeping glaze stone into glaze ash; the more the glaze ash, the lower meltability the glaze and the smoother the surface of the burned glazed surface. But too much glaze ash would make the glaze surface dark-green and grey while the smaller glaze ash would make the glaze surface pure white but easy to be spotted.

DIVISION OF LABOUR BASED ON SPECIALISATION There were various types of work in the key links of production

including shaping, firing, and colour painting. The shaping process was divided into two major professions of "rounding ware (*yuanqi*)" and "engraving ware (*zuoqi*)" according to manufacturing methods. "Rounding ware" means to model the body on the potting wheel, which was half-dried after it was thrown from the mud hump, with which those porcelain goods for daily use such as bowls, plates, cups and dishes were produced. "Engraving ware" means to turn the rough shape of the body, which had been thrown beforehand on the potting wheel, into a finish body through the fine trimming on both inner and outer sides of the porcelain, by which products of vases, pots, jars, *Gang*, *Bo* and so on were produced. Among the professions of rounding ware and engraving ware, their varieties were also divided into throwing, modelling, carving, spinning, inlaying, drilling, arching, paring, engraving (openworking), etc. Taking the shaping in rounding ware as an example: "charring (*daza*)" means to compose all kinds of raw materials and wash them into mud; "body-making (*zuopei*)" means to put the mud lump onto spinning potting wheel and keep two hands on the mud and pull the mud into body through the in-and-out and the back-and-forth of hands and fingers; "body-modelling (*yinpei*)" means to press and pat the thrown body on the model to make it even and uniform in the size of the model; "body-trimming (*lipei*)" means to whirl and pare the semi-finished body so as to make its surface bright and clean; "bottom-handle-removing (*guapei*)" means to make the body bottom flat and clean for incising of hallmark; "body-glazing (*shahepei*)" means to glaze the first layer after the first time of trimming body and the second layer after the second time of trimming body.

During the firing stage, the firing technology was divided into the kiln keeping (*bazhuang*), the *tuopei*, the sagger placing (*zhuangxia*), the glazing and colourizing (*jiabiao*), the cutting bottom-handle (*shoudoujiao*), the charring (*daza*), the assisting (*xiaofushou*) and so on. The person in charge of the kiln keeping was of great importance since he should be looking over the strength of fire and control the firing.

The division of drawing *Qinghua*: The painter drew the outline of patterns on the body and the inker stuffed the light greenish blue into the lines which was called as *fenshui*. The *Qinghua* string pattern was made by the body spinner (body-throwing-maker) and all bottom hallmarks were written by a specially-designated person.

The division of colour painting on glaze: The painter drew the patterns and scenery on the product with mono-colour pigment while the paint-stuffer (colour-filling maker) applied the paints to